Research Gaps and Project Plan

LLM Hallucination Detection via Knowledge Augmentation

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March 2026

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1 Overview

This document identifies the key research gaps left open by existing literature on LLM hallucination detection, and outlines the plan for addressing them in this thesis through a knowledge-grounded symbolic verification approach.

The central research question driving this thesis is:

> *How can symbolic domain knowledge be used to detect logically or factually inconsistent LLM outputs within a restricted domain?*

2 Research Gaps

The following five gaps were identified after a systematic review of the related literature covering zero-resource detection methods, external knowledge-based detection, constrained decoding, and knowledge augmentation approaches.

2.1 Gap 1 — Unstructured vs. Structured Knowledge References

What the literature does: Existing external knowledge-based approaches such as FActScore [4] and REFCHECKER [5] verify LLM outputs against unstructured text sources like Wikipedia, relying on natural language understanding for matching, which introduces ambiguity and imprecision in the verification process. GraphEval [10] partially addresses this by constructing a knowledge graph from LLM outputs and checking extracted triples for consistency. However, GraphEval still relies on an NLI model to match triples against unstructured contextual text rather than a structured symbolic knowledge base, meaning the ambiguity of natural language interpretation is not eliminated but merely shifted to a different stage of the pipeline.

What is missing: No existing approach uses a structured symbolic knowledge graph as the reference for fully deterministic, direct triple-to-triple matching that requires no NLI model or natural language understanding at any stage. Structured symbolic graphs would eliminate ambiguity entirely and allow exact constraint checking.

How this thesis addresses it: This thesis uses a hand-crafted symbolic knowledge graph within a restricted domain as the verification reference, enabling fully deterministic matching without any natural language interpretation or NLI model at any stage of the verification pipeline.

2.2 Gap 2 — Zero-Resource Methods Fail on Consistent Hallucinations

What the literature does: Consistency-based approaches such as SelfCheckGPT [3], FINCH-ZK [6], GCA [7], FactSelfCheck [8], and Lie to Me [9] detect hallucinations by measuring consistency across multiple LLM samples. If responses are consistent, the claim is assumed to be factual.

What is missing: If an LLM confidently generates and repeatedly reproduces a false claim across all samples, consistency-based methods will incorrectly label it as factual.

No approach uses an external symbolic source to detect this failure mode.

How this thesis addresses it: By checking outputs against an explicit external knowledge graph, the verification layer catches hallucinations regardless of how consistently the LLM repeats them.

2.3 Gap 3 — No Work Targets Rule-Violating Hallucinations in Restricted Domains

What the literature does: All surveyed detection approaches — both zero-resource and external knowledge-based — target open-domain factual correctness. They evaluate whether LLM claims match real-world facts across unrestricted topics.

What is missing: No work specifically focuses on detecting hallucinations that violate explicit, formalizable domain rules such as loan eligibility conditions, budget constraints, or game move legality. This is a distinct and unaddressed category of hallucination.

How this thesis addresses it: The project targets a deliberately restricted domain with clearly defined symbolic rules. Hallucinations are detected specifically when LLM outputs violate those domain constraints.

2.4 Gap 4 — No Domain-Specific Hallucination Taxonomy

What the literature does: Existing taxonomies such as the intrinsic/extrinsic classification by Ji et al. [1] and the factuality/faithfulness categorization by Huang et al. [2] are designed for general-purpose, open-domain settings.

What is missing: No taxonomy exists specifically for constraint-violation hallucinations within a restricted symbolic domain. Categories such as rule violations, missing required conditions, and fabricated domain facts are not captured by existing frameworks.

How this thesis addresses it: A domain-specific hallucination taxonomy will be defined, covering categories including rule violations, logical inconsistencies, missing required conditions, and fabricated domain facts.

2.5 Gap 5 — No Systematic Comparison of LLM Alone vs. LLM + Symbolic Verification

What the literature does: Papers such as FActScore and REFCHECKER evaluate detection performance but do not isolate the contribution of symbolic verification by comparing it against an unaugmented LLM baseline in a controlled domain setting.

What is missing: No study provides a clean, controlled experimental comparison measuring exactly how much a symbolic verification layer reduces undetected rule violations versus using the LLM alone.

How this thesis addresses it: The evaluation will systematically compare two conditions: (1) LLM alone and (2) LLM with the symbolic verification layer, measuring violation detection rate, types of hallucinations caught, and false positives.

3 Research Plan

The following plan maps directly onto the six steps described in the project guide, with estimated timelines and deliverables for each step.

3.1 Step 1 — Select a Restricted Domain

Goal: Commit to one domain with well-defined, unambiguous, symbolically formalizable rules.

Candidate domains:

- **Finance** — loan eligibility rules and budget constraints
- **Policy/Eligibility** — rule-based approvals and conditions
- **Games** — legal moves and scoring rules
- **University Rules** — course prerequisites and credit requirements

Criteria for selection: The domain must have fewer than 20 core rules, allow full symbolic formalization, and avoid any ambiguity in rule interpretation.

Deliverable: A written domain specification listing all rules that will be encoded.

3.2 Step 2 — Formalize Domain Knowledge

Goal: Represent all domain rules as a structured knowledge graph using triples of the form $\langle subject, predicate, object \rangle$.

Tasks:

- List all domain rules in natural language
- Convert each rule into one or more symbolic triples
- Encode the knowledge graph (e.g., using NetworkX or a simple JSON structure)
- Validate completeness and consistency of the graph

Deliverable: A fully encoded domain knowledge graph with documented rules.

3.3 Step 3 — Generate LLM Outputs

Goal: Design prompts that elicit multi-step domain reasoning from a black-box LLM, producing structured outputs that are amenable to verification.

Tasks:

- Design at least 50 prompts covering the full rule set
- Prompt the LLM to produce structured, step-by-step reasoning
- Collect and store outputs in a consistent format
- Manually annotate a subset as ground truth for evaluation

Deliverable: A dataset of LLM outputs with ground truth hallucination labels.

3.4 Step 4 — Implement the Verification Layer

Goal: Build a symbolic reasoning component that parses LLM outputs, extracts claims as triples, and checks them against the knowledge graph.

Tasks:

- Extract claim triples from LLM outputs (using prompt-based parsing)
- Query the knowledge graph to check each extracted triple
- Flag violations with the specific constraint broken
- Keep the parser simple and the checking logic deterministic

Deliverable: A working end-to-end verification pipeline.

3.5 Step 5 — Define the Hallucination Taxonomy

Goal: Categorize detected hallucinations into a structured taxonomy relevant to the chosen domain.

Proposed categories:

- **Rule Violation** — output explicitly contradicts a domain rule
- **Logical Inconsistency** — output contains internal contradictions
- **Missing Required Condition** — output omits a mandatory constraint
- **Fabricated Domain Fact** — output invents entities or values not in the knowledge graph

Deliverable: A documented taxonomy with examples from the generated dataset.

3.6 Step 6 — Evaluation

Goal: Systematically compare LLM alone versus LLM with the symbolic verification layer.

Metrics:

- Violation detection rate (recall)
- False positive rate (precision)
- Breakdown of detected hallucinations by taxonomy category

Experimental conditions:

1. **Baseline:** LLM output evaluated by human annotators only
2. **Proposed:** LLM output evaluated by the symbolic verification layer

Deliverable: An empirical results table and analysis section in the thesis.

4 Timeline

Week	Step	Task
1	Domain Selection + Formalization	Finalize domain, document all rules, build and validate knowledge graph
2	LLM Output Generation	Design prompts, collect outputs, manually annotate ground truth labels
3–4	Verification Layer	Implement triple extractor and symbolic constraint checker
5	Taxonomy + Evaluation	Categorize hallucinations, run LLM-alone vs LLM+verification experiments
6	Writing	Complete methodology, results, and conclusion chapters

Critical Constraints for 6-Week Execution

Given the compressed timeline, the following scope decisions are essential:

- **Keep the domain tiny** — aim for 10–15 rules maximum. More rules means more formalization time and more prompts to annotate.
- **Keep the dataset small but sufficient** — 30–50 annotated LLM outputs is enough for a controlled evaluation at bachelor level.
- **Use a simple triple extractor** — prompt the LLM itself to extract triples from its own output rather than building a custom NLP parser.
- **Hardcode the knowledge graph** — use a Python dictionary or JSON file rather than a full graph database. NetworkX is acceptable if needed.
- **No custom models** — the verification layer must be purely rule-based and deterministic. No training, no fine-tuning.
- **Writing runs in parallel** — start drafting the methodology chapter in Week 3 as implementation progresses, not after it finishes.

5 Summary of Contributions

This thesis makes the following contributions that together address all five identified gaps:

1. A restricted-domain symbolic knowledge graph used as a deterministic verification reference (addresses Gap 1)
2. An external symbolic verification layer that catches consistent hallucinations missed by zero-resource methods (addresses Gap 2)
3. The first detection system specifically targeting rule-violating hallucinations in a restricted symbolic domain (addresses Gap 3)
4. A domain-specific hallucination taxonomy for constraint-violation errors (addresses Gap 4)

5. A controlled empirical comparison of LLM alone versus LLM with symbolic verification (addresses Gap 5)

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